

Value of plasma D-dimer level for prediction of in-hospital mortality in patients presenting with Takotsubo syndrome

Hiroki Mochizuki ^{*}, Tsutomu Yoshikawa, Konomi Sakata, Tetsuo Yamaguchi , Toshiaki Isogai, Yoichi Imori , Kenshiro Arao, Yoshimitsu Takaoka, Yukihiro Watanabe, Toru Egashira, Toshiaki Otsuka, Takeshi Yamamoto, Ken Nagao, Shun Kohsaka, and Morimasa Takayama 

Tokyo CCU Network Scientific Committee, Tokyo, Japan

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Aims

D-dimer, degradation product of cross-linked fibrin, has been described as the prognostic marker in some cardiovascular diseases. However, the association between D-dimer levels and prognosis has not been fully evaluated in patients with Takotsubo syndrome (TTS).

Methods and results

We retrospectively analysed data of 580 patients with TTS from the Tokyo Cardiovascular Care Unit Network registry. The primary endpoint was in-hospital all-cause death. Logistic regression analysis was used to investigate the relationship between D-dimer level and mortality. The additive effect of D-dimer level on the conventional prognostic risk score (InterTAK) was assessed using C-statistic, net reclassification improvement (NRI), and integrated discrimination improvement (IDI). In the entire sample (median age, 77 years; 79.5% women), 174 (30.0%) patients had an emotional trigger, while 177 (30.5%) patients had a physical trigger, causing 28 (4.8%) deaths (cardiac death [$n = 13$, 46.4%], non-cardiac death [$n = 15$, 53.6%]). D-dimer levels on admission were significantly higher in those who died than in those who survived (6.4 [3.9–18.3] vs. 1.1 [0.7–2.9] $\mu\text{g/mL}$, $P < 0.001$). On multivariable logistic regression analysis, the odds ratio (OR) for in-hospital mortality significantly increased with D-dimer levels $\geq 3.5 \mu\text{g/mL}$ (OR: 7.06 [95% confidence interval: 2.90–17.16], $P < 0.001$). The InterTAK Prognostic Score was 17 (11–24), and NRI and IDI were improved by incorporating D-dimer levels (NRI: 1.08, $P < 0.001$; IDI: 0.05, $P < 0.001$).

Conclusion

Elevated D-dimer levels may serve as an additional predictor of increased in-hospital mortality in patients with TTS. Patients with higher D-dimer levels may warrant intensified monitoring and management.

* Corresponding author. Tel: +81-3-3541-5151, Fax: +81-3-3549-0699, Email: hmochizuki0430@gmail.com

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Graphical Abstract

Value of Plasma D-dimer Level for Prediction of In-Hospital Mortality in Patients Presenting with Takotsubo Syndrome

Study design

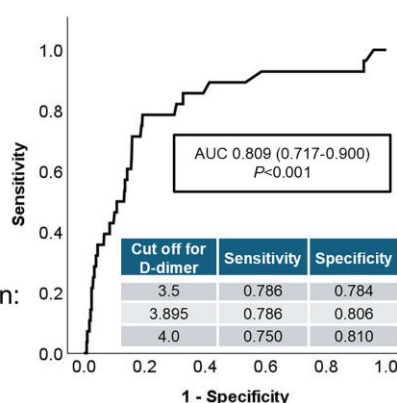
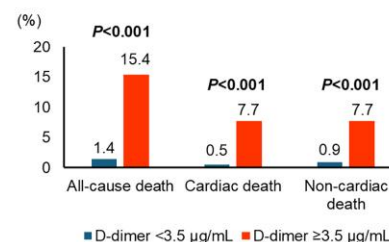
- Retrospective, Multicenter, Observational registry
- Data from the Tokyo Cardiovascular Care Unit Network registry
- 76 major hospitals in Tokyo were enrolled.

Population



- 580 patients with takotsubo syndrome in Tokyo 2013-2017
- Median age: 77 years
- 79.5% woman
- Median D-dimer level on admission: **1.2 µg/mL**
- **4.8%** in-hospital all-cause death

ROC curve for D-dimer level to predict all-cause death

In-hospital mortality
D-dimer level ≥ 3.5 µg/mL

In-hospital all-cause death
OR 7.06
(95% CI 2.90-17.16)

ROC, receiver operating characteristic; AUC, area under the curve; OR, odds ratio; CI, confidence interval

Keywords

Takotsubo syndrome • D-dimer • In-hospital mortality • Cardiac death • Non-cardiac death

Introduction

Takotsubo syndrome (TTS) is an acute cardiovascular syndrome characterized by reversible left ventricular (LV) dysfunction.¹ To date, the exact pathophysiology of this condition remains unclear. Several mechanisms have been proposed, including catecholamine-induced myocardial stunning, multivessel epicardial spasm, acute microvascular dysfunction, inflammatory mechanisms and metabolic dysregulation.^{2,3}

D-dimer levels are indicative of fibrin production and degradation, thereby reflecting the turnover of the coagulation system.⁴ D-dimer levels can be measured immediately in the clinical setting and are widely used in the diagnosis of venous thromboembolism.⁵ It has been also studied as a prognostic marker in some cardiovascular diseases associated with activation of the coagulation and, such as myocardial infarction, acute aortic syndrome, and heart failure.⁶⁻⁸ The activation of coagulation cascade is also caused by the catecholamine excess states.⁹ Moreover, inflammation promotes coagulation by upregulating pro-coagulant factors and inhibiting fibrinolytic activity.¹⁰ However, the association between D-dimer levels and the prognosis of patients with TTS has not been fully evaluated. The present study aimed to investigate

the relationship between D-dimer levels and in-hospital mortality in patients with TTS.

Methods

Study design

We used patient data from the Tokyo Cardiovascular Care Unit (CCU) Network registry, a multicentre, retrospective cohort registry comprising data on all patients hospitalized in the CCU. The Tokyo CCU Network was founded in 1978 and has now enrolled 76 major hospitals in Tokyo. The details of the Tokyo CCU Network have been previously described.^{11,12} The Takotsubo cardiomyopathy study group was established in 2010. Diagnoses were made by the attending physicians at each hospital, using the revised Mayo Clinic criteria, namely, the presence of transient LV wall motion abnormality beyond a single epicardial coronary artery perfusion territory, the absence of obstructive coronary artery disease or angiographic evidence of acute plaque rupture, the presence of new electrocardiogram abnormalities (either ST-segment elevation and/or T-wave inversion) or elevation in cardiac troponin levels, and the absence of pheochromocytoma and myocarditis.¹³ Data on D-dimer levels have been included in the registry since January 2013; therefore, data on patients

hospitalized with TTS from January 2013 to December 2017 were used for the current study. Patients who did not undergo coronary angiography and those whose D-dimer level data were not available were excluded from the present analyses. The requirement for informed consent was waived because all the data were anonymized. The current study was approved by the institutional review board of the Tokyo CCU Network Scientific Committee (approval no. 116-002) and conformed to the ethical principles of the Declaration of Helsinki.

Definitions of outcomes

The primary endpoint was in-hospital all-cause death, including cardiac death and non-cardiac death. Cardiac death was defined as death due to cardiogenic shock, heart failure, arrhythmia, or myocardial infarction. Non-cardiac death was defined as death other than cardiac death, such as that due to sepsis or malignancy.

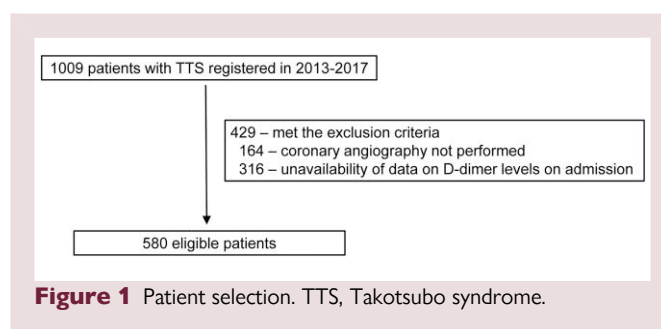
Statistical analysis

Among the baseline characteristics of patients, categorical variables are reported as frequencies with proportions and compared using Pearson's χ^2 test, while continuous variables are reported as medians with quartiles and compared using the Mann-Whitney *U* test. Kruskal-Wallis test and the Bonferroni correction were used for examining statistical differences among three groups. Further, a receiver operating characteristic (ROC) curve and a precision recall (PR) curve were drawn to predict the primary outcome using D-dimer levels. The optimal D-dimer cut-off point was evaluated using the Youden index from ROC curve. To reduce the small-sample bias, we performed multivariable logistic regression analysis based on Firth's penalized likelihood method for in-hospital all-cause mortality, adjusting for variables with $P < 0.10$ in the univariable analysis. The selection of variables for univariate analysis was based on clinical importance and findings from previous studies.^{1,14–20} The following were selected: age, sex, diabetes mellitus, trigger factors, systolic blood pressure on admission, heart rate on admission, white blood cell (WBC) count, C-reactive protein (CRP), creatinine (Cr), B-type natriuretic peptide (BNP), and left ventricular ejection fraction (LVEF). WBC, CRP, Cr, and BNP data were converted into binary variables based on the median values. Missing values were imputed using multiple imputations before multivariable regression analysis. To assess the predictive ability of D-dimer, we incorporated D-dimer into the InterTAK Prognostic Score and calculated the area under the ROC curve (AUC) for discrimination, category-free net reclassification (NRI), and integrated discrimination improvement (IDI). We compared the AUC between the InterTAK Prognostic Score and a modified InterTAK Prognostic Score using DeLong's test. The InterTAK Prognostic Score was obtained from the International Takotsubo (InterTAK) registry,¹ which can stratify patients at risk for all-cause mortality both in the short and long term according to their clinical profiles, comprising age, sex, trigger factors, diabetes mellitus, heart rate, systolic blood pressure, and LVEF.²¹ The InterTAK Prognostic Score was categorized into four risk groups: low risk ≤ 15 points, intermediate risk 16–22 points, high risk 23–28 points, and very high risk ≥ 29 points. Following the method of original article using bootstrapping to develop coefficient-based scoring system, we added points if the D-dimer level was above the cut-off point of the InterTAK Prognostic Score. A two-sided $P < 0.05$ was considered statistically significant. Statistical analysis was performed using SPSS version 27 (IBM Corporation, Armonk, NY, USA) and R, version 4.5.0 (The R Foundation for Statistical Computing, Vienna, Austria).

Results

Baseline characteristics

A total of 1009 patients hospitalized with TTS between 2013 and 2017 were identified in the Tokyo CCU Network registry. After excluding patients with pre-defined exclusion criteria, no performance of coronary angiography ($n = 164$, 16.3%) and missing data on D-dimer ($n = 316$, 31.3%), 580 patients were finally included (Figure 1). The baseline characteristics are shown in Table 1. The median age was 77 years (interquartile range [IQR]: 68–82 years), and 79.5% of the patients were women. In the entire sample, 174 patients (30.0%) had an emotional trigger, while 177 patients (30.5%) had a physical trigger; the



other patients had no or unidentifiable triggers. The median D-dimer level was 1.20 $\mu\text{g/mL}$ (IQR: 0.70–3.34 $\mu\text{g/mL}$). Anticoagulant medicines, such as heparin and warfarin, were used in 464 out of 580 patients (80.0%) to prevent thrombosis. We also reperformed the analysis with the excluded patients and found that the baseline characteristics and outcomes were mostly not different between the two groups (see Supplementary File S1). During hospitalization, 28 out of 580 patients (4.8%) died. There was no significant relationship between mortality and the balloon type, whereas patients who died had a higher physical stress ratio. Regarding laboratory findings, patients who died had higher WBC counts and CRP levels.

Outcomes

The plasma D-dimer levels in patients who died were significantly elevated than in those who survived (6.4 $\mu\text{g/mL}$ [IQR: 3.9–18.3 $\mu\text{g/mL}$] vs. 1.1 $\mu\text{g/mL}$ [IQR: 0.7–2.9 $\mu\text{g/mL}$], $P < 0.001$). Among the patients who died, there was no significant difference about D-dimer levels between patients with emotional triggers and patients with physical triggers (8.6 $\mu\text{g/mL}$ [IQR: 0.9–16.3 $\mu\text{g/mL}$] vs. 9.4 $\mu\text{g/mL}$ [IQR: 4.8–22.9 $\mu\text{g/mL}$], $P = 0.665$), although there was significant difference between patients with physical triggers and patients with no or unidentifiable triggers (9.4 $\mu\text{g/mL}$ [IQR: 4.8–22.9 $\mu\text{g/mL}$] vs. 3.1 $\mu\text{g/mL}$ [IQR: 0.9–5.3 $\mu\text{g/mL}$], Bonferroni-adjusted $P = 0.019$) (Figure 2). Moreover, there was no significant difference about D-dimer levels between patients with emotional triggers, each subtype of physical trigger, and no or unidentified triggers (data not shown). The rates of cardiac death and non-cardiac death were 46.4% (13 of 28 patients) and 53.6% (15 of 28 patients), respectively. The causes of cardiac death included heart failure ($n = 10$, 76.9%), arrhythmia ($n = 1$, 7.7%), myocardial infarction ($n = 1$, 7.7%), and others ($n = 1$, 7.7%). The causes of non-cardiac death included pneumonia ($n = 2$, 13.3%), acute cholangitis ($n = 1$, 6.7%), sepsis ($n = 3$, 20.0%), subarachnoid haemorrhage ($n = 1$, 6.7%), brain herniation ($n = 1$, 6.7%), malignancy ($n = 3$, 20.0%), and others ($n = 4$, 26.7%). On ROC analysis, D-dimer levels had a high predictive value for in-hospital all-cause mortality (AUC 0.809, 95% confidence interval [CI] 0.717–0.900, $P < 0.001$) (Figure 3). D-dimer levels also had a high predictive value for cardiac death (AUC 0.803, 95% CI 0.666–0.940, $P < 0.001$) and non-cardiac death (AUC 0.799, 95% CI 0.679–0.920, $P < 0.001$) (see Supplementary File S2). On the other hand, on PR analysis, D-dimer levels had a low positive predictive value for all-cause death (AUC 0.191, 95% CI 0.060–0.390), cardiac death (AUC 0.061, 95% CI 0.008–0.153), and non-cardiac death (AUC 0.171, 95% CI 0.029–0.447) (see Supplementary File S3). For all-cause death, the optimal D-dimer cut-off value from ROC analysis was found to be 3.895 $\mu\text{g/mL}$ (Youden index 0.592), with a sensitivity of 78.6% and specificity of 80.6%. Youden indexes were 0.570 at 3.5 $\mu\text{g/mL}$ and 0.560 at 4.0 $\mu\text{g/mL}$, and the present cut-off point was set to 3.5 $\mu\text{g/mL}$, considering the clinical setting. The high D-dimer group ($\geq 3.5 \mu\text{g/mL}$) had higher in-hospital all-cause death, cardiac death, and non-cardiac death (Figure 4). Table 2 shows the univariable, and multivariable logistic

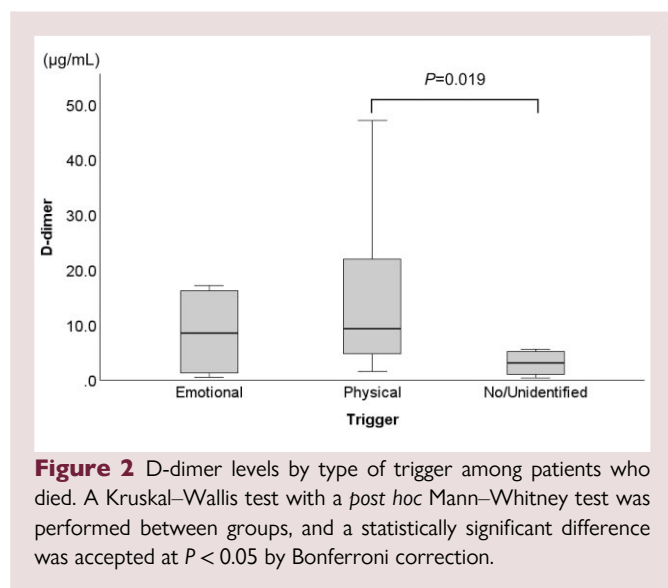
Table 1 Patient characteristics

	Total (n = 580)	Patients who survived (n = 552)	Patients who died (n = 28)	P value
Age, years	77 (68–82)	77 (68–82)	80 (73–85)	0.012
Sex: female	461 (79.5)	444 (80.4)	17 (60.7)	0.027
Body mass index, kg/m ²	20.4 (18.2–23.2) (n = 552)	20.4 (18.3–23.2) (n = 524)	20.3 (16.6–22.4) (n = 24)	0.584
Prior history				
Myocardial infarction	7/466 (1.2)	6/445 (1.3)	1/21 (4.8)	0.277
Coronary revascularization	13/466 (2.2)	12/445 (2.7)	1/21 (4.8)	0.455
Heart failure hospitalization	9/466 (1.6)	9/445 (2.0)	0/21 (0.0)	1.000
Stroke	38/466 (6.6)	38/445 (8.5)	0/21 (0.0)	0.401
Comorbidities				
Hypertension	212/466 (36.6)	202/445 (45.4)	10/21 (47.6)	1.000
Dyslipidaemia	108/466 (18.6)	104/445 (23.4)	4/21 (19.0)	0.795
Diabetes mellitus	79/466 (13.6)	76/445 (17.1)	3/21 (14.3)	1.000
Pre-admission medication				
ACE-I/ARB	142/558 (24.5)	135/532 (25.4)	7/26 (26.9)	0.820
β -blocker	55/558 (9.5)	53/532 (10.0)	2/26 (7.7)	1.000
Calcium channel blocker	165/557 (28.4)	155/531 (29.2)	10/26 (38.5)	0.378
Balloon types				0.323
Apical	522/576 (90.0)	496/548 (90.5)	26/28 (92.9)	
Mid-ventricular	37/576 (6.4)	36/548 (6.6)	1/28 (3.6)	
Base	12/576 (2.1)	12/548 (2.2)	0/28 (0.0)	
Triggering event				<0.001
Emotional	174/572 (30.0)	170/544 (31.3)	4/28 (14.3)	
Physical	177/572 (30.5)	159/544 (29.2)	18/28 (64.3)	
Neurologic disorder	22/177 (12.4)	18/159 (11.3)	4/18 (22.2)	
Infection	48/177 (27.1)	45/159 (28.3)	3/18 (16.7)	
Malignancy	13/177 (7.3)	12/159 (7.5)	1/18 (5.6)	
Other medical conditions	50/177 (28.2)	47/159 (29.6)	3/18 (16.7)	
Surgery or trauma	32/177 (18.1)	25/159 (15.7)	7/18 (38.9)	
Physical activities	12/177 (6.8)	12/159 (7.5)	0/18 (0.0)	
Unidentifiable	221/572 (38.1)	215/544 (39.5)	6/28 (21.4)	
Vital signs				
Body temperature, °C	36.5 (36.1–36.8) (n = 544)	36.5 (36.1–36.8) (n = 517)	36.6 (36.0–37.7) (n = 27)	0.210
Systolic blood pressure, mmHg	130 (111–152) (n = 574)	131 (112–152) (n = 547)	121 (104–162) (n = 27)	0.393
Heart rate, beats/min	87 (74–100) (n = 570)	86 (74–100) (n = 542)	95 (73–117) (n = 28)	0.131
Respiratory rate, times/min	19 (16–24) (n = 477)	19 (16–24) (n = 457)	22 (18–28) (n = 20)	0.100
Blood test findings				
WBC, / μ L	8580 (6300–11 200) (n = 579)	8400 (6200–11 100) (n = 551)	10 400 (7573–14 125) (n = 28)	0.011
CRP, mg/dL	0.30 (0.10–2.44) (n = 571)	0.30 (0.10–2.09) (n = 543)	4.37 (0.99–14.56) (n = 28)	<0.001
Cr, mg/dL	0.72 (0.56–0.97) (n = 580)	0.71 (0.56–0.96) (n = 552)	0.94 (0.69–1.77) (n = 28)	0.016
CK, U/L	153 (86–289) (n = 578)	153 (86–279) (n = 550)	150 (80–414) (n = 28)	0.723
BNP factor increase in ULN, pg/mL	10.8 (3.2–29.2) (n = 496)	10.3 (3.0–28.5) (n = 474)	34.4 (8.5–99.1) (n = 22)	0.001
D-dimer, μ g/mL	1.2 (0.70–3.34) (n = 580)	1.1 (0.7–2.9) (n = 552)	6.4 (3.9–18.3) (n = 28)	<0.001
ST elevation, n (%)	383/577 (66.0)	360/549 (65.6)	23/28 (82.1)	0.099
LVEF, %	53.0 (42.1–62.0) (n = 452)	53.0 (43.0–62.0) (n = 434)	36.7 (30.0–59.0) (n = 18)	0.013
In-hospital complications				
VF or sustained VT	18/577 (3.1)	12/550 (2.2)	6/27 (22.2)	<0.001
Complete AVB	7/577 (1.2)	6/550 (1.1)	1/27 (3.7)	0.286
Length of hospital stay, days	10 (6–17)	10 (6–17)	10 (2–33)	0.484

Categorical variables are shown as numbers (n [%]) or n/total n (%), while continuous variables are represented as median (interquartile range).

BNP factor increase in the ULN was defined as the ratio of BNP or N-terminal pro-brain natriuretic peptide levels to the ULN.

ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; WBC, white blood cell; CRP, C-reactive protein; Cr, creatinine; CK, creatine kinase; BNP, brain natriuretic peptide; ULN, upper limit of the normal range; LVEF, left ventricular ejection fraction; VF, ventricular fibrillation; VT, ventricular tachycardia; AVB, atrioventricular block.

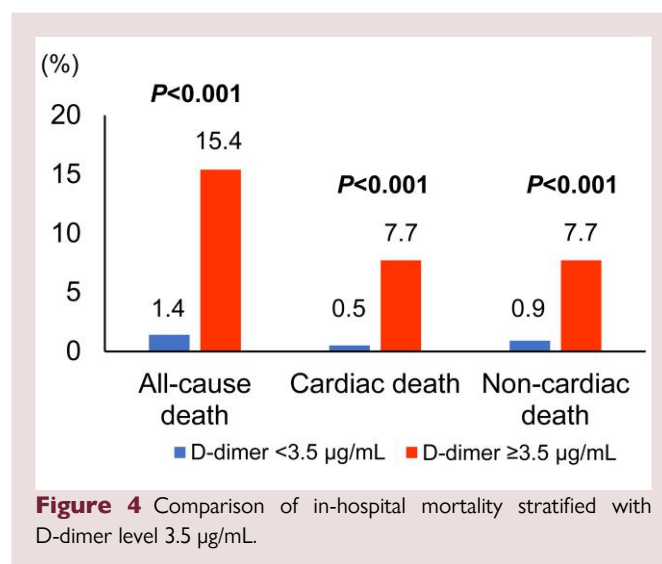
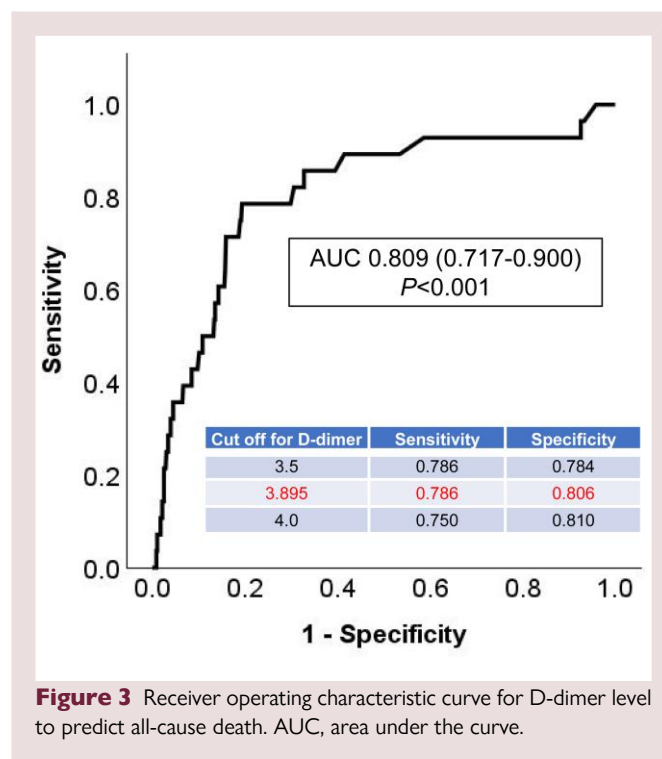


regression analyses based on Firth's penalized likelihood method for in-hospital all-cause mortality. The adjusted odds ratio (OR) for in-hospital mortality significantly increased with higher D-dimer levels ≥ 3.5 $\mu\text{g/mL}$ (OR, 7.06 [95% CI, 2.90–17.16]; $P < 0.001$). These results did not differ from those from using a dataset without the multiple imputation (see [Supplementary File S4](#)). Higher D-dimer levels ≥ 3.5 $\mu\text{g/mL}$ were also an independent predictor for both cardiac death (OR, 9.47 [95% CI, 2.73–32.86]; $P < 0.001$) and non-cardiac death (OR, 4.15 [95% CI, 1.43–12.04]; $P = 0.009$) (see [Supplementary File S5](#)). The InterTAK Prognostic Score was 17 (IQR: 11–24), which was an intermediate risk in the present study. We added 15 points to a D-dimer levels ≥ 3.5 $\mu\text{g/mL}$ according to the methodology using bootstrapping to develop coefficient-based scoring system. [Figure 5](#) shows the ROC analysis for the prediction of all-cause mortality after adding D-dimer to the InterTAK prognostic score, and AUC increased from 0.792 to 0.841 although it was not statistically significant ($P = 0.148$). On the other hand, The NRI ([Table 3](#)) and IDI were improved by incorporating D-dimer into the InterTAK prognostic score (NRI: 1.08 [95% CI, 0.62–1.52], $P < 0.001$; IDI: 0.05 [95% CI, 0.02–0.07], $P < 0.001$).

Discussion

D-dimer is a degradation product of cross-linked fibrin, and higher D-dimer levels imply the existence of a hypercoagulable state and thrombus formation and degradation. The impact of D-dimer levels on the prognosis of patients with cardiovascular diseases has been described. Menown *et al.* reported that elevated D-dimer levels were predictive of death and myocardial infarction in patients who presented with chest pain but had normal levels of troponin.⁶ D-dimer levels have also been associated with poor outcomes in patients with systolic heart failure.⁸ In the present study, we found that D-dimer levels on admission were significantly higher among patients with TTS who died during hospitalization. This finding suggests that a higher D-dimer level is a predictor of higher in-hospital mortality in patients with TTS.

Several hypotheses have been proposed to explain the cardiac response to severe stress in TTS, and a catecholamine surge induced by emotional or physical stress is one of the recent hypotheses. In vitro study demonstrated high-dose epinephrine could induce a switch in beta-adrenoreceptor coupling from the Gs to Gi protein, leading to prevent excessive activation of the myocyte and reduce contractility.²² Catecholamine excess is generally known to cause thrombotic



tendencies due to various mechanisms including hypercoagulable states, endothelial damage, blood stasis, and platelet aggregation.⁹ Cecchi and colleagues reported that von Willebrand factor (vWF) and plasminogen activator inhibitor-1 (PAI-1) levels in endothelial cells, which were endothelial damage biomarkers, were significantly higher in patients with TTS.^{23–25} Moreover, they observed that patients with TTS had higher values of blood viscosity and erythrocyte aggregation. However, regarding thrombus formation and D-dimer levels, different results have been reported. A previous small study showed D-dimer was elevated in 60% of 15 patients with TTS.²⁶ Contrastingly, Parkkonen *et al.* compared haematological parameters among 47 patients with TTS and equal age- and sex-matched patients with acute coronary syndrome and those without coronary artery disease.²⁷

Table 2 Univariable and multivariable logistic regression analysis for all-cause death

	Univariable analysis			Multivariable analysis (n = 580)		
	Odds ratio	95% CI	P value	Odds ratio	95% CI	P value
D-dimer, ≥ 3.5 $\mu\text{g/mL}$	13.06	5.18–32.94	<0.001	7.06	2.90–17.16	<0.001
Age ≥ 75 years	1.58	0.70–3.55	0.271			
Sex: male (Ref: female)	2.66	1.21–5.84	0.015	1.61	0.72–3.58	0.248
BMI $\leq$ median (20.4 kg/m^2)	1.00	0.44–2.27	1.000			
Diabetes mellitus	0.81	0.23–2.81	0.736			
Triggering event (Ref: Emotional)						
Emotional		Ref			Ref	
Physical	5.12	1.70–15.45	0.004	1.54	0.52–4.57	0.435
Unidentifiable	1.20	0.33–4.33	0.778	0.73	0.22–2.44	0.608
Systolic blood pressure ≤ 90 mmHg	1.33	0.30–5.88	0.705			
Heart rate > 100 beats/min	2.51	1.17–5.41	0.019	1.49	0.68–3.26	0.313
WBC $\geq$ median (8580/ μL)	2.18	0.97–4.91	0.059	1.10	0.48–2.50	0.818
CRP $\geq$ median (0.30 mg/dL)	23.21	3.13–172.02	0.002	2.04	0.12–34.53	0.620
Cr $\geq$ median (0.72 mg/dL)	2.56	1.11–5.90	0.028	1.23	0.53–2.88	0.625
CK $\geq$ median (152 U/L)	1.00	0.47–2.12	0.985			
BNP factor increase in ULN $\geq$ median (10.8 pg/mL)	2.78	1.07–7.23	0.036	1.36	0.57–3.21	0.488
LVEF $\leq 45\%$	3.34	1.27–8.79	0.015	2.69	1.24–5.84	0.012

BMI, body mass index; WBC, white blood cell; CRP, C-reactive protein; Cr, creatinine; CK, creatine kinase; BNP, brain natriuretic peptide; ULN, upper limit of the normal range; LVEF, left ventricular ejection fraction; CI, confidence interval; Ref, reference.

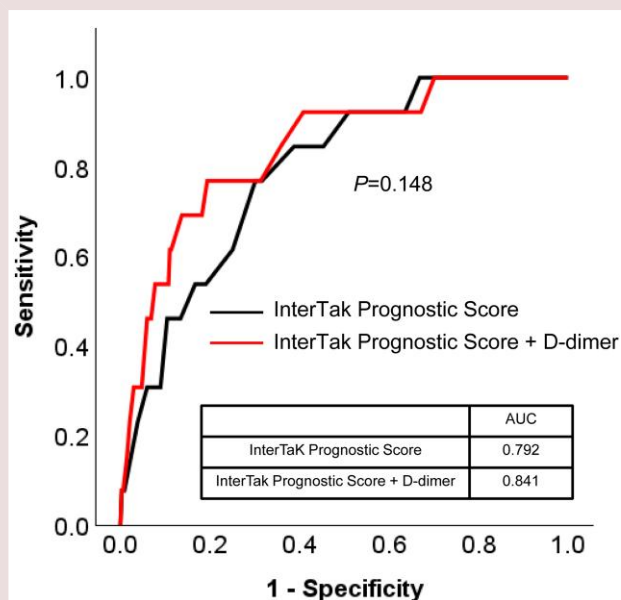


Figure 5 Comparison of receiver operating characteristic curve for adding D-dimer level to InterTAK Prognostic Score. AUC, area under the curve.

Although coagulation factors such as fibrinogen and vWF increased accordingly, D-dimer was elevated only in 30% of patients with TTS, similar to that in patients without coronary artery disease and less commonly than that in patients with acute coronary syndrome.

Cecchi *et al.* also reported that patients with TTS showed D-dimer values similar to control groups and lower than patients with acute coronary syndrome.²⁴ Scholars have also proposed that transient intracoronary thrombotic occlusion is involved in the pathophysiology of TTS; however, a recent study using intravascular ultrasound showed no evidence of plaque rupture, positive remodelling, or presumed intracoronary thrombus in patients with TTS.²⁸ Accruing these results, although it appears that catecholamines such as epinephrine and norepinephrine induced by stress may cause endothelial damage (mediated by an increase in PAI-1 and vWF levels) and blood hyperviscosity, consequently leading to acute microvascular hypoperfusion, which may be one of the causes of TTS, thrombus formation may not be a direct cause of TTS.

In the present study, patients who died had more physical triggers, particularly infection, malignancy, and surgery/trauma. These conditions are known to be associated with coagulation abnormalities, and elevated D-dimer levels have also been associated with the prognosis and severity of these diseases.^{29–32} Moreover, in our cohort, most cardiac deaths were due to heart failure. Patients with heart failure experience a certain degree of hypercoagulability, secondary to blood stasis, inflammation, neurohormonal activation, and endothelial dysfunction; accordingly, the association between elevated D-dimer levels and poor prognosis has been reported in other cardiovascular disease.^{8,33,34}

Myocardial inflammation is suspected to be another main cause of TTS in previous studies. Wilson HM *et al.* investigated the effect of catecholamine on myocardial inflammation in a rat model.³⁵ They showed early neutrophil infiltrates, followed by type 1 macrophages after the 3rd day. Recently, the same groups demonstrated a myocardial macrophage inflammatory infiltrate using cardiac magnetic resonance imaging, including ultrasmall superparamagnetic particles of iron oxide (USPIO) enhancement for the detection of inflammatory macrophages.³⁶ They also revealed an increase in systemic inflammatory cytokines/chemokines such as interleukin-6 (IL-6), interleukin-8 (IL-8), and chemokine (C-X-C motif) ligand (CXCL) 1. IL-8

Table 3 Reclassification table using category-free NRI. Model with D-dimer in addition to InterTAK Prognostic Score

	All	Reclassified upwards <i>n</i> (proportion)	Reclassified downwards <i>n</i> (proportion)	NRI	
Number of patients who died	13	10 (0.77)	3 (0.23)	Among cases	0.54 (0.08–1.00)
Number of patients who were alive	335	76 (0.23)	259 (0.77)	Among controls	0.55 (0.45–0.63)
Overall NRI (CI)					1.08 (0.62–1.52)

NRI, net reclassification improvement; CI, confidence interval.

and CXCL1 induce monocyte adhesion and macrophage infiltration or release of inflammatory cells from the bone marrow.³⁷ IL-6 is a proinflammatory marker, and another group found that several factors, including epinephrine, induce IL-6 mRNA expression from cardiac myocytes using a rat model of hypoxic myocardial stress.³⁸ IL-6 also plays an important role in the activation of coagulation.³⁹ Therefore, the chronic inflammatory state may be enhanced by catecholamine surges and lead to the activation of coagulation. In our registry, Murakami *et al.* previously reported that a higher WBC count was associated with in-hospital cardiac complications.¹⁸ Arao *et al.* also reported that there was an association between high CRP levels and in-hospital all-cause death.¹⁹

On ROC analysis, the best cut-off point of the D-dimer level was 3.5 µg/mL for in-hospital all-cause death. This suggests that relatively high D-dimer levels in clinical settings may reflect severe myocardial damage in patients with TTS. D-dimer can be measured immediately and easily in the emergency department, unlike other coagulation activation markers and inflammatory markers, and we should be cautious when the D-dimer level is above 3.5 µg/mL because of increased risk of complications. Moreover, we demonstrated that adding D-dimer levels to the InterTAK Prognostic Score improved the risk reclassification and discrimination for in-hospital all-cause death. D-dimer levels may be used as a part of an algorithm to assess the in-hospital prognosis of TTS patients.

Study limitations

The present study has some limitations. First, this was a retrospective, observational study, and unmeasured variables may have influenced despite covariate adjustment. Also, the dataset used in this study was limited to the period during (2013–2017). The registry underpinning the present analysis was established with Institutional Review Board approval in 2010 and prospectively collected data on D-dimer levels between January 2013 and December 2017, corresponding to the principal investigator's original funding period and data-use agreement. After 2017, case report forms and additional ethics approvals would have been required to merge the newer data; moreover, the onset of the COVID-19 pandemic introduced substantial changes in admission patterns and diagnostic protocols that could confound temporal trend analyses. To preserve methodological consistency and ensure complete adjudication of endpoints with the variables originally specified, we therefore confined the dataset to the 2013–2017 enrolment window. Second, due to restricted study period and some missing data, the number of in-hospital deaths may have been too small for multivariable analysis and discrimination ability. Third, there are several commercially available D-dimer assays, such as latex agglutination assays, immunofluorescent assays, and enzyme-linked immunosorbent assays (ELISA).⁴⁰ Latex-enhanced turbidimetric immunoassay is widely used in Japan, but D-dimer assays were not unified in the hospitals enrolled in this study. Fourth, the time from trigger onset to blood sampling could not be accurately assessed. Unmeasured variation in sampling times may have introduced additional noise into the D-dimer analyses and should be addressed in future prospective studies. Fifth, in the present study, intraventricular thrombi were not fully evaluated. We sometimes encounter patients who have intraventricular

thrombus, leading to reduced LVEF. We need to rethink the prognostic value of D-dimer levels in TTS patients who have intraventricular thrombus because the coagulation system may accelerate. Anticoagulant medicines, such as heparin and warfarin, were used in most patients to prevent thrombosis. Although anticoagulant medicines may influence serum D-dimer levels, blood tests may be checked before admission in most patients and D-dimer levels are probably not influenced by use of anticoagulant medicines. Finally, we only had data on in-hospital outcomes, and the clinical impact of D-dimer levels on long-term outcomes is unclear.

Conclusions

The present study showed that plasma D-dimer levels were significantly higher in patients who died during hospitalization. A higher D-dimer level may be predictive of higher in-hospital mortality in patients with TTS. Moreover, adding D-dimer levels to the existing risk scoring system (InterTAK Prognostic Score) improved the risk stratification for outcome. Also, TTS patients with markedly elevated D-dimer should undergo definitive imaging to exclude left-ventricular thrombus and, when detected, receive guideline-directed anticoagulation.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

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Author contributions

Hiroki Mochizuki: Conceptualization, Data curation, Foraml analysis, Methodology, Visualization, Writing—original draft. Tsutomu Yoshikawa: Writing—review & editing, Supervision. Konomi Sakata: Writing—review & editing. Tetsuo Yamaguchi: Writing—review & editing. Toshiaki Isogai: Writing—review & editing. Yoichi Imori: Writing—review & editing. Kenshiro Arao: Writing—review & editing. Yoshimitsu Takaoka: Writing—review & editing. Yukihiro Watanabe: Writing—review & editing. Toshiaki Otsuka: Methodology, Writing—review & editing. Toru Egashira: Writing—review & editing. Takeshi Yamamoto: Writing—review & editing. Ken Nagao: Writing—review

& editing. Shun Kohsaka: Writing—review & editing. Morimasa Takayama: Writing—review & editing.

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Data availability

Because of ethical restrictions imposed by the institutional review board committee, data are only available to members of the Tokyo CCU Network. Therefore, the Tokyo CCU Network cannot provide individual raw data to third parties.

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